

Server Room Requirements & Setup Tasks

AI Face Detection & Event Photo Platform — 30 TB In-House Deployment

Ticket Ref: SR-TICKET-2026-01

Target Infrastructure: 30 TB Allocated NAS

Priority: High (Production Rollout)

Date: September 2026

Executive Objective: Provision a lightweight host VM, mount our allocated 30 TB NAS storage pool, launch the containerized AI Face Detection service via Docker Compose, and route incoming guest traffic securely through Cloudflare Tunnel.

1 TASK 1: Host Virtual Machine (VM) Provisioning

Please provision a virtual machine (or container host) with the following specifications:

Resource	Recommended Specification	Notes & Details
Operating System	Ubuntu 22.04 LTS or 24.04 LTS	Debian 12 is also fully supported.
CPU Allocation	4 to 8 Cores (x86_64)	Parallel face detection tensor inference.
Memory (RAM)	8 GB to 16 GB RAM	InsightFace AI model + Qdrant in-memory cache.
System Disk (OS)	30 GB to 50 GB SSD	For OS, Docker images, and vector database index.
Software Engine	Docker Engine & Docker Compose	Install command: <code>curl -fsSL https://get.docker.com sudo sh</code>

2 TASK 2: 30 TB NAS Storage Mount & Photographer Network Share

1. Mount NAS on the Host VM: Mount our allocated 30 TB NAS photo volume to the VM at:

```
# Mount point directory:
/mnt/storage/event_photos

# Example NFS mount command:
sudo mount -t nfs -o rw,hard,intr 192.168.1.50:/volume1/event_photos /mnt/storage/event_photos

# Set permissions for container read/write:
sudo chmod -R 775 /mnt/storage/event_photos
```

2. Create Office Network Share (SMB / CIFS):

- Expose this same folder on the local office network as a Windows/SMB Network Share: `\<NAS-IP>\event_photos`
- Workflow:** Our camera crew will connect their laptops to the office Wi-Fi/LAN and drag-and-drop raw SD card photo folders (e.g., `\192.168.1.50\event_photos\6-10-Grand-Gala\`) directly into this NAS share. No cloud uploads needed!

Application Execution & Network Ingress

Docker Compose Configuration & Cloudflare Subdomain Routing

Host Port: 8000 (Internal)
Repository: jasirmjr/Face-Detection
SSL / TLS: Cloudflare Edge

3 TASK 3: Deploy Face Detection Application (Docker Compose)

On the host VM, create the application directory, save the following docker-compose.yml file, and start the service:

```
# 1. Create project directory
mkdir -p /opt/face-finder && cd /opt/face-finder

# 2. Save docker-compose.yml:
version: '3.8'

services:
  face-finder:
    image: python:3.11-slim
    container_name: face_finder
    restart: unless-stopped
    ports:
      - "8000:8000"
    volumes:
      # Mount the 30TB NAS folder where camera crew drops event albums:
      - /mnt/storage/event_photos:/app/storage
      # Persistent vector database for facial feature vectors:
      - ./qdrant_storage:/app/qdrant_storage
    working_dir: /app
    environment:
      - PORT=8000
      - PYTHONUNBUFFERED=1
    command: >
      bash -c "apt-get update && apt-get install -y git build-essential ffmpeg libsm6 libxext6 &&
        git clone https://github.com/jasirmjr/Face-Detection.git /tmp/repo &&
        cp -r /tmp/repo/* /app/ &&
        pip install --no-cache-dir -r /app/requirements.txt &&
        python /app/run.py"

  deploy:
    resources:
      limits:
        cpus: '8.0'
        memory: 16G

# 3. Launch service
docker compose up -d

# 4. Verify health check
curl http://localhost:8000/health
# Output: {"status":"ok","engine_ready":true}
```

4 TASK 4: Public HTTPS Exposure via Cloudflare Tunnel

Event venue guests will search for their photos from their mobile phones (using 4G/5G mobile data):

- Route our company subdomain (e.g., photos.ourcompany.com or events.ourcompany.com) through Cloudflare Tunnel to the host at http://localhost:8000.
- Security Advantage:** Zero inbound router ports need to be opened. Cloudflare handles automatic HTTPS (SSL certificate) and DDoS protection.

✓ Operator Handover Details (Please Complete & Return to Project Lead)

Local Network Share UNC: \ _____ \event_photos
Public Guest URL: https:// _____

Server Operator Name:

____/____/2026

Date: